JAVA BACKEND

Interview Questions by Company

248 questions across 9 interview experiences, in the order they were asked

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COMPANY 01

32 QUESTIONS

Capgemini

Java Backend Interview

Covered Java 8 & 17, Spring Boot, Microservices, Kafka, SonarQube, WebSocket and WebClient.

CORE JAVA & JAVA 8/17

10 Qs

1	What are the important features introduced in Java 17?
2	Why upgrade from Java 8 to Java 17, even if we don't use features like Records or Sealed Classes?
3	What are the advantages of Java 17 over Java 8?
4	What are the major features introduced in Java 8?
5	What is a default method in an interface? Why was it introduced?
6	What is a static method in Java, and why is it useful?
7	What is the Stream API?
8	What is a method reference?
9	Given a list of Person objects containing name and age, sort by age in ascending order using Java 8.
10	What is the difference between map() and flatMap()?

SPRING BOOT & MICROSERVICES

8 Qs

11	Why do we use Spring Boot?
12	What is auto-configuration in Spring Boot?
13	What is @SpringBootApplication?
14	What is @ComponentScan?
15	What is a Circuit Breaker in Microservices?
16	What is Service Discovery, and why is it needed?
17	How do you register a microservice with a Service Discovery server?
18	If Service A calls Service B, how does the complete Service Discovery flow work?

KAFKA

6 Qs

19	Have you worked with Kafka?
20	What is Spring Kafka, and why is it used?
21	What is a Kafka Consumer?
22	What is a Consumer Group?

23	How are Kafka partitions assigned to consumers within a Consumer Group?
24	Can the same Kafka partition be consumed by consumers from different Consumer Groups?

CODE QUALITY & MODERN JAVA

5 Qs

25	Have you worked with SonarQube and code coverage?
26	How do you implement code coverage in an application?
27	What is the purpose of code coverage?
28	Why do we use SonarQube?
29	From a performance and security perspective, why are applications moving from Java 8 to Java 17?

API & COMMUNICATION

3 Qs

30	What is WebSocket?
31	What is WebClient?
32	What is the difference between RestTemplate and WebClient?

COMPANY 02 75 QUESTIONS

Honeywell

Java Interview (shared by Jashu Kumar)

Prepare across Core Java, Spring Boot, microservices, SQL, concurrency and coding. The post ends with "For Honeywell specifically, also prepare project-based questions:" - that list was not visible in the screenshots.

CORE JAVA

15 Qs

1	What are the main OOP principles in Java?
2	Difference between == and .equals().
3	Why is String immutable?
4	Difference between String, StringBuilder, and StringBuffer.
5	How does HashMap work internally?
6	Difference between HashMap, ConcurrentHashMap, and Hashtable.
7	What happens when two keys have the same hash code?
8	Explain equals() and hashCode() contract.
9	Difference between ArrayList and LinkedList.
10	Difference between final, finally, and finalize.
11	Checked vs unchecked exceptions.
12	What is try-with-resources?
13	Abstract class vs interface.
14	What are Java records?
15	What is dependency injection?

MULTITHREADING & JAVA 8+

15 Qs

16	Process vs thread.
17	synchronized vs Lock.
18	What is a deadlock? How do you prevent it?
19	volatile vs AtomicInteger.
20	What is ExecutorService?
21	submit() vs execute().
22	Callable vs Runnable.
23	What is CompletableFuture?
24	Synchronous vs asynchronous execution.

25	Explain Java Stream API.
26	map() vs flatMap().
27	filter() vs map().
28	Intermediate vs terminal stream operations.
29	What is a functional interface?
30	What is Optional and why use it?

SPRING BOOT

10 Qs

31	What is Spring Boot?
32	@Component vs @Service vs @Repository.
33	@Controller vs @RestController.
34	Constructor injection vs field injection.
35	What is Spring Boot auto-configuration?
36	What is Spring Boot Actuator?
37	How do you handle exceptions globally?
38	Explain @Transactional.
39	What happens if a transaction fails?
40	How do you configure multiple databases?

MICROSERVICES

14 Qs

41	What are microservices?
42	Monolith vs microservices.
43	How do services communicate?
44	REST vs messaging.
45	What is API Gateway?
46	What is service discovery?
47	How does Resilience4j work?
48	Explain Circuit Breaker.
49	Retry vs Circuit Breaker.
50	How would you handle distributed transactions?
51	What is Saga pattern?
52	How do you secure microservices using JWT?
53	How do you handle inter-service failures?
54	How do you monitor microservices?

SQL / DATABASE

10 Qs

55	Find the second-highest salary.
56	Difference between INNER JOIN and LEFT JOIN.
57	What is an index?
58	When can an index make performance worse?
59	How do you optimize a slow SQL query?
60	What is database connection pooling?
61	What is the N+1 query problem?
62	Explain database normalization.
63	Optimistic vs pessimistic locking.
64	What are ACID properties?

CODING QUESTIONS

11 Qs

65	Reverse a string without using built-in reverse.
66	Find duplicate elements in an array.
67	Find the first non-repeating character.
68	Check whether two strings are anagrams.
69	Find the missing number from 1..N.
70	Find the longest substring without repeating characters.
71	Two Sum.
72	Merge two sorted arrays.
73	Implement an LRU cache.
74	Find frequency of each word using Java Streams.
75	Sort employees by salary using Streams.

COMPANY 03 12 QUESTIONS

Wissen

Java Backend | 13 YOE

Focused on Java, Kafka, Microservices, Concurrency, Transactions, Production Troubleshooting and System Design.

KEY QUESTIONS

12 Qs

1	How would you design an immutable class in Java?
2	Implement a custom Iterator and a custom Exception.
3	How do you decide the number of Kafka partitions?
4	Design a system that guarantees per-user event ordering while maintaining high throughput.
5	Using Java Streams, process transaction data and find the Top 3 clients by transaction amount.
6	Analyze a multithreading scenario involving synchronized instance vs static methods.
7	Production API latency increased from 200ms to 5 seconds. How would you troubleshoot it?
8	How would you handle Kafka poison messages, retries, DLT/DLQ and offsets?
9	Explain Saga, Transactional Outbox, CDC and Distributed Locking.
10	How would you design Kafka-based credit/debit processing with ordering and high throughput?
11	Explain HashMap vs ConcurrentHashMap vs LinkedHashMap.
12	Design a Distributed Log Aggregation System.

Key areas: Java | Kafka | Microservices | Concurrency | Database | System Design | Production Support

COMPANY 04 19 QUESTIONS

EPAM

Java Developer Interview

Covered DSA, Java Internals, Spring Boot, Multithreading, Security and Microservices.

CODING & JAVA STREAMS 3 Qs

1	DSA: Arrange elements around a given threshold - smaller elements first, larger elements after.
2	Java Streams: Convert "I love Java" to "#ILoveJava".
3	Streams: Find anagrams using Java Streams.

CORE JAVA 9 Qs

4	HashMap & HashSet internals
5	Treeification
6	wait() vs sleep() vs yield()
7	Thread vs Runnable
8	Thread & Multithreading
9	ReentrantLock vs synchronized
10	Java Memory Model (JMM)
11	Garbage Collection basics
12	Design a Thread Pool

SPRING BOOT & SECURITY 5 Qs

13	@Configuration, @Service, @SpringBootApplication - output-based questions
14	How do Spring Boot annotations work internally?
15	Global Exception Handling
16	Spring Security
17	Authentication vs Authorization

MICROSERVICES 2 Qs

18	Microservices Design
19	Designing scalable and reliable services

COMPANY 05 20 QUESTIONS

Production Scenarios

Java First Round 2026 (company not named)

> *"The Java first round changed in 2026. It is no longer just a coding test. It is 20 production scenarios." The post does not name a company.*

CORE JAVA 5 Qs

1	HashMap has 1 million entries. Performance is degrading. Why, and how would you resolve it?
2	Your code uses synchronized everywhere. The application is slow. What is causing the bottleneck?
3	Double-checked locking in a singleton. Why is volatile required?
4	A ThreadLocal variable is causing a memory leak. How does it happen?
5	wait vs sleep. How can each contribute to deadlock scenarios?

JVM AND MEMORY 3 Qs

6	The application ran fine for 30 days, then crashed with OutOfMemoryError. No code changes. What happened?
7	GC runs every 30 seconds and pauses the application for 2 seconds. How would you optimize it?
8	When would you choose G1GC over ZGC in a production environment?

SPRING BOOT 4 Qs

9	@Transactional is applied to a private method. Why does it silently have no effect?
10	Two beans have the same type. How does Spring determine which one to inject?
11	An @Async method throws an exception. Nobody catches it. What happens?
12	Application startup takes 45 seconds. How would you diagnose and optimize it?

DATABASE AND JPA 3 Qs

13	LazyInitializationException occurs in production but not in development. Why?
14	A JPA query performs well at 100 rows but takes 45 seconds at 1 million. What is wrong?
15	Two transactions update the same row simultaneously. One silently overwrites the other. How would you prevent it?

SYSTEM DESIGN AND PRODUCTION 5 Qs

16	Your payment API receives the same request twice due to a network retry. How do you prevent a double charge?

17	Service A calls B. B is down. A keeps retrying, preventing B from recovering. What pattern fixes this?
18	Redis caching is working, but database load remains at 100%. What could be wrong?
19	Response time is 20ms normally but reaches 8 seconds under load. No errors appear in logs. Where do you start?
20	A feature was deployed Friday evening. By Monday morning, everything is slow. Nobody changed the code. What happened?

COMPANY 06 17 QUESTIONS

Deloitte

Java Developer - Round 1 Technical (shared by Vihan R.)

Round 1 - Technical Interview.

CORE JAVA & JAVA 8

7 Qs

1	What are the key features introduced in Java 8?
2	Difference between map() and flatMap().
3	What are intermediate and terminal operations in Stream API?
4	Functional interface vs marker interface.
5	Can we overload the main() method? Which one does JVM execute?
6	Difference between this and super keywords.
7	How do you create and handle custom exceptions?

MULTITHREADING & CONCURRENCY

6 Qs

8	Difference between Runnable and Callable.
9	Difference between start() and run().
10	Difference between wait() and sleep().
11	What is synchronization in Java?
12	Explain Java memory areas: Heap, Stack, Method Area, and Program Counter.
13	What is garbage collection? What is the purpose of finalize()?

CODING & SQL

4 Qs

14	Find the minimum-length subarray whose sum equals a given target.
15	Write an SQL query to rank employees based on their experience.
16	Difference between RANK(), DENSE_RANK(), and ROW_NUMBER().
17	Explain the order of execution of SQL query clauses.

COMPANY 07 29 QUESTIONS

Coforge

Senior Java / Spring Boot (shared by Anubhi Sharma)

Covered Kafka, Spring Boot, REST APIs, databases, Java 17/21, caching, system design and behavioral scenarios.

KAFKA & EVENT-DRIVEN ARCHITECTURE 6 Qs

1	How would you monitor whether Kafka consumers are keeping up with the message backlog?
2	What metrics and alerts would you configure for consumer backlog?
3	After identifying the cause of a backlog, what would you do next?
4	How does Kafka handle message retention?
5	What happens when a message exceeds its retention period?
6	How can you change topic retention settings?

DATABASE & REST API 5 Qs

7	Is PostgreSQL SQL or NoSQL?
8	What are the different REST API versioning strategies?
9	How would you version an API without breaking existing clients?
10	Pros/cons of URI, query-parameter, header-based, and media-type versioning?
11	What is InnoDB in MySQL?

SPRING BOOT SECURITY 4 Qs

12	How would you securely manage authentication and authorization for web and mobile clients?
13	How would you implement role-based and token-based access control?
14	How would you implement method-level authorization using @PreAuthorize?
15	How do you securely handle passwords in Spring Boot?

HIGH-THROUGHPUT SYSTEMS 3 Qs

16	How would you design a Spring Boot application for high throughput and real-time event processing?
17	How would you ensure consistency, fault tolerance, and low latency?
18	What components and design patterns would you use?

REST API + JPA

1 Qs

19	Design a product API supporting: Category filtering; Price sorting; Pagination; JPA Specification/Criteria filtering; Invalid-parameter exception handling.

JAVA CACHING & CONCURRENCY

4 Qs

20	How would you implement a high-performance Java cache?
21	Which Java 17/21 features or APIs would you leverage?
22	How would you handle cache eviction?
23	How would you handle concurrency?

JAVA 17/21 MODERNIZATION

3 Qs

24	How would you identify legacy classes suitable for conversion to Java 17 records?
25	Benefits and challenges of records in business models?
26	When would you NOT use records, especially with JPA/Hibernate entities?

BEHAVIORAL & STAKEHOLDER MANAGEMENT

3 Qs

27	How would you facilitate consensus when senior stakeholders have conflicting viewpoints?
28	How would you ensure all perspectives are respected during verbal communication?
29	If choosing one person over three other candidates/stakeholders, how would you explain the choice without appearing biased?

COMPANY 08 29 QUESTIONS

Optum

Java Interview Questions I Was Asked

Deep, internals-focused Java, concurrency, Spring and production-debugging questions.

ALL QUESTIONS

29 Qs

1	How does intern() work with the String pool? What's the memory-level difference between new String("abc") and "abc"?
2	How would you implement immutability in Java? Why is String immutable - explain the internal reasoning.
3	How is memory allocated for static, final, and static final variables in the JVM?
4	How does the diamond problem get resolved in Java through default methods in interfaces?
5	Why aren't multiple abstract methods allowed in functional interfaces?
6	What performance issues can autoboxing/unboxing cause in production?
7	How does the var keyword (Java 10) resolve type inference internally at compile time?
8	What is the correct use case for the Optional class, and how is it commonly misused?
9	How does the Reflection API work in Java, and what are its performance trade-offs?
10	Give a real production example where the Liskov Substitution Principle was violated.
11	Describe a scenario where using inheritance instead of composition broke the design.
12	How did the internal structure of HashMap change after Java 8 (treeify threshold, red-black tree)?
13	How does ConcurrentHashMap internally handle locking (segment locking vs CAS)?
14	What is the performance impact during HashMap resize/rehashing, and how can it be avoided?
15	When should CopyOnWriteArrayList be used, and what is its memory overhead?
16	What is the use case for WeakHashMap and IdentityHashMap?
17	Explain the difference between ForkJoinPool and a normal ExecutorService in the context of work-stealing.
18	How does ThreadLocal create a memory leak, and how do you avoid it?
19	Explain the practical difference between thenApply(), thenCompose(), and thenCombine() in CompletableFuture.
20	Explain the use case for Semaphore in the context of connection pooling.
21	What is the happens-before relationship in the Java Memory Model?
22	How does exception chaining (cause) help with debugging in production?
23	Explain all phases of the Spring Bean lifecycle, up to BeanPostProcessor.

24	How does Spring resolve circular dependencies, and when does it fail?
25	How does the Spring Security filter chain internally process a request?
26	How do you use Spring Boot Actuator for production monitoring?
27	Explain how the G1 Garbage Collector works and why it's better than CMS for high-heap applications.
28	What is the Hibernate N+1 query problem, and how do you detect/fix it in production?
29	How do you detect a memory leak in the JVM (through heap dump analysis tools)?

COMPANY 09 15 QUESTIONS

Flipkart

Java Developer - Executor Framework & Thread Pools

Executor Framework and thread-pool questions, plus ExecutorService revision topics (2-5 YOE roles).

EXECUTOR FRAMEWORK QUESTIONS

15 Qs

1	What is the Executor Framework in Java?
2	What is the difference between Executor and ExecutorService?
3	What is the difference between execute() and submit()?
4	What is a ThreadPoolExecutor?
5	What is the difference between Runnable and Callable?
6	What is the purpose of the Future interface?
7	What is the difference between shutdown() and shutdownNow()?
8	What happens if all worker threads are busy?
9	What is a thread pool queue?
10	What is a RejectedExecutionHandler?
11	How do you handle exceptions in submitted tasks?
12	What is the difference between Future and CompletableFuture?
13	How do you configure a thread pool for a Spring Boot application?
14	How do you prevent thread pool exhaustion in production?
15	When would you use ExecutorService instead of creating threads manually?

EXECUTORSERVICE REVISION TOPICS

10 topics

- What is ExecutorService?
- Why is ExecutorService better than creating Threads manually?
- execute() vs submit()
- Runnable vs Callable
- Future interface
- shutdown() vs shutdownNow()
- Different types of thread pools
- Why newCachedThreadPool() is dangerous
- Why ThreadPoolExecutor is preferred in production
- Real-world project usage examples